Lab 1: record sound and explain the data

- What the microphone and recorder measure
- Choose sampling rate and duration; recognize aliasing
- Record three sounds, export MAT files, inspect and replay
- Practice the same measurement decisions in the prepared pre-lab

Control the variables you can

- Independent variable: the setting deliberately changed, such as sample rate.
- Dependent result: the measured waveform, RMS, or FFT peak.
- Extraneous influences: room noise, distance, orientation, and strike strength.
- Change one setting at a time; record what you could not hold constant.

What will the three sounds look like?

- Tuning fork: a nearly sinusoidal voltage with a decaying amplitude.
- Whistle: a sustained tone that may drift and include harmonics.
- Finger snap: a short transient containing a range of frequencies.
- Background sound and electrical noise also enter the recorded signal.

From air pressure to stored numbers

SOUND		MICROPHONE		ADC		FILE
Pressure $p(t)$, Pa	→	Analog voltage $x(t)$, V	→	Sample + quantize	→	Time and voltage samples

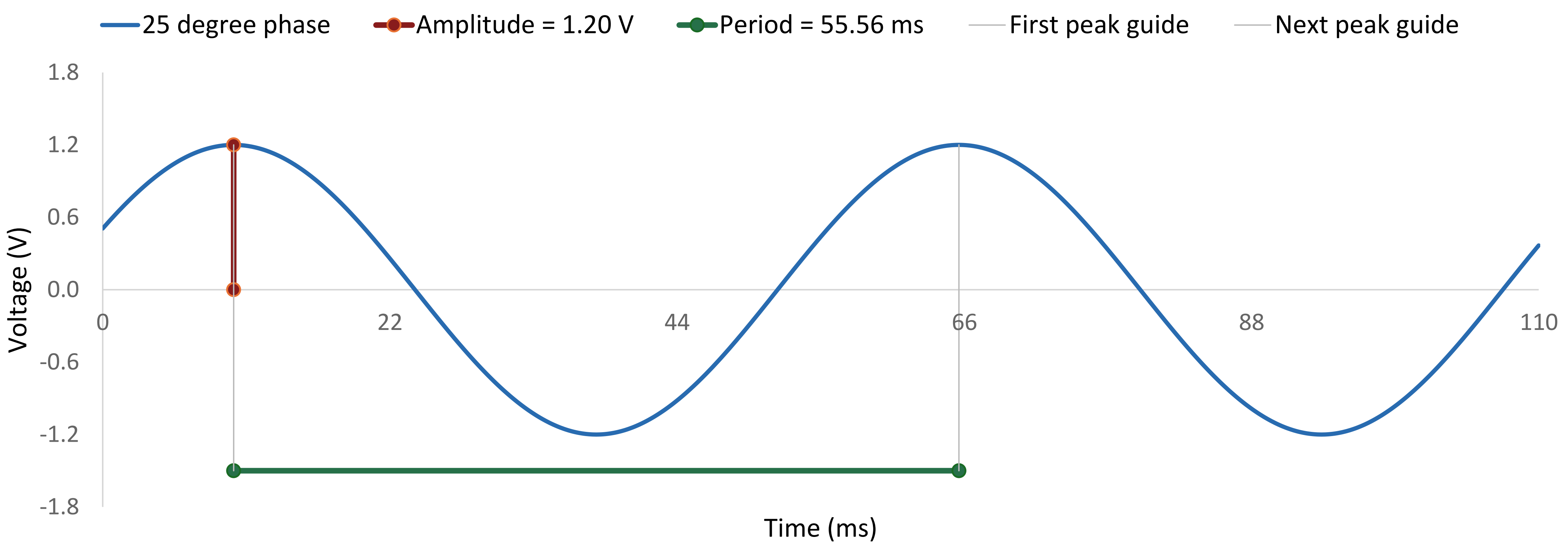
Sampling: $x[n] = x(n\Delta t)$ selects the measurement times.

Sample-and-hold: hold a voltage steady while it is converted.

Quantization: round to a finite set of voltage levels.

Time spacing and voltage resolution are different limitations.

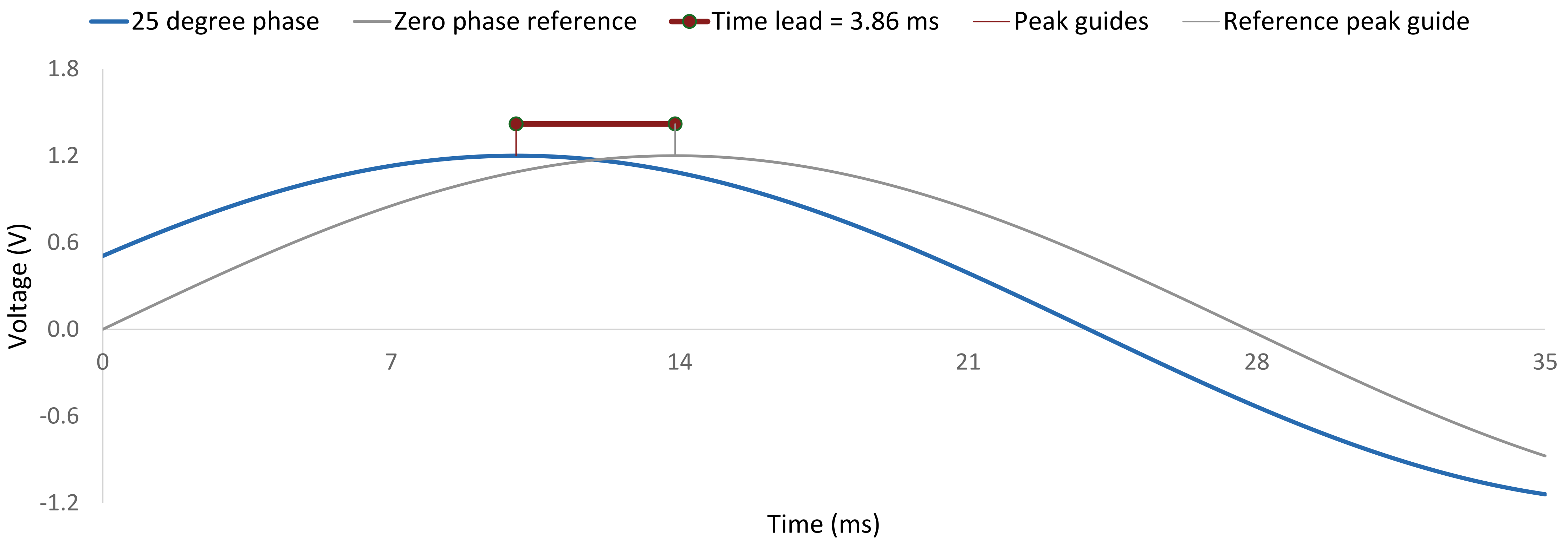
A sinusoid: what the three parameters mean



A_1 = 1.20 V from zero to peak. T_1 = 55.56 ms between consecutive peaks.

$x_1(t) = A_1 \sin(2\pi f_1 t + \phi_1)$, with f_1 = 18 Hz and ϕ_1 = 25°

Phase relative to a zero-phase sine wave



ϕ_1 = +25°: the blue wave reaches its peak 3.86 ms earlier

Time lead = ϕ_1 / (360° f_1). Positive phase shifts a sine wave to the left.

A teaching example with two frequency components

- $x(t) = 1.20 \sin(2\pi \cdot 18t + 25°) + 0.45 \cos(2\pi \cdot 42t - 15°)$ V
- $A1 = 1.20$ V; $f1 = 18$ Hz; phi1_deg = 25° for tone 1.
- $A2 = 0.45$ V; $f2 = 42$ Hz; phi2_deg = −15° for tone 2.
- f_max = 42 Hz even though this component has the smaller amplitude.

Mean and RMS: continuous and sampled

- Mean: $\mu = (1/T) \int x(t)\, dt \rightarrow \mu = (1/N) \Sigma\, x[n]$
- RMS: $x\_rms = \sqrt{[(1/T) \int x^2(t)\, dt]} \rightarrow \sqrt{[(1/N) \Sigma\, x^2[n]]}$
- Over whole cycles: a zero-offset sinusoid has $\mu = 0$ and $RMS = A/\sqrt{2}$.
- With offset: $RMS^2 = \mu^2 + RMS^2$ of the zero-mean component.

Signal frequency and sampling rate

THE WAVE: f_signal

f_signal counts complete cycles per second.

Unit: hertz (Hz).

Our signal has two components:

f_1 = 18 Hz and f_2 = 42 Hz.

THE RECORDER: f_s

f_s counts stored samples per second.

Unit: samples/s (often written Hz).

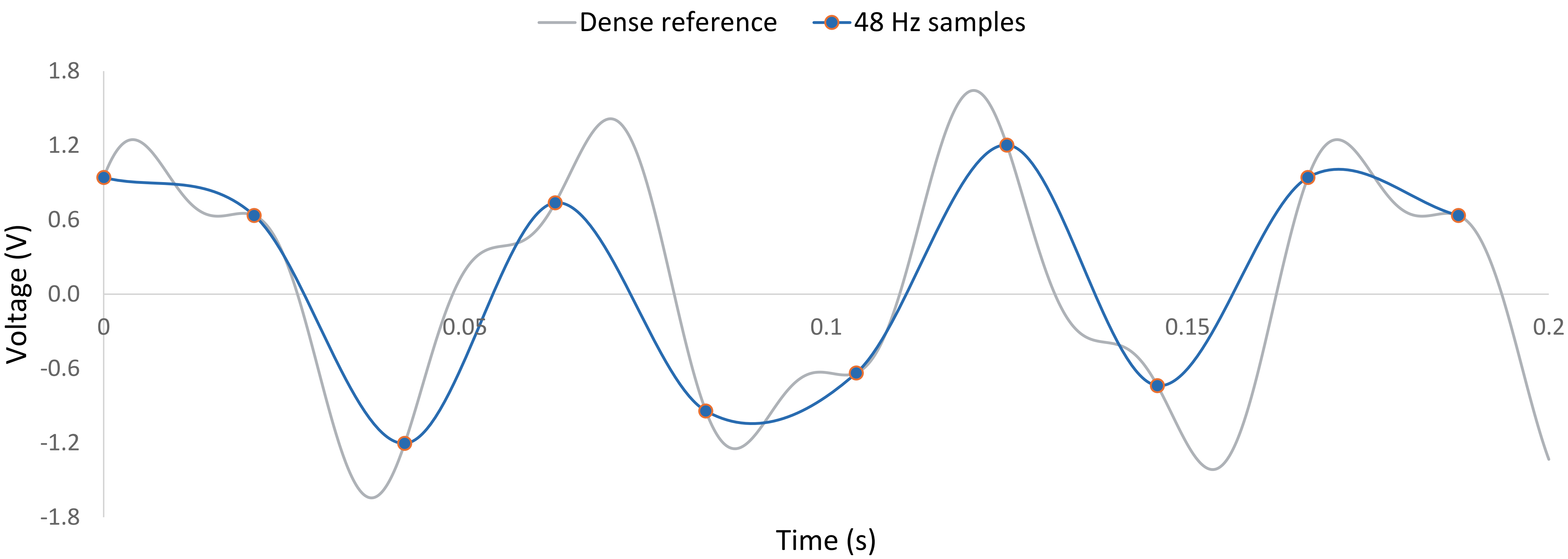
We choose this setting on the recorder.

Changing f_s changes the sample times.

Sampling interval and record duration

- $\Delta t = 1/f_s$ is the time between adjacent samples.
- N is the number of stored samples; $T_{record} = N/f_s$.
- Sample times: $0, \Delta t, \ldots, (N - 1)\Delta t$; do not duplicate the endpoint.
- Example: $f_s = 1000$ samples/s and $N = 2000$ give a 2 s record.

Samples and the underlying waveform



Markers are simulated samples of the known signal. Connecting lines add no measurements.

The Nyquist condition

THE SIGNAL REQUIREMENT

f_{max} is the highest frequency present.

For our two tones: $f_{max} = 42$ Hz.

To avoid aliasing in this signal:

$f_s > 2 f_{max} = 84$ samples/s.

THE RECORDER LIMIT

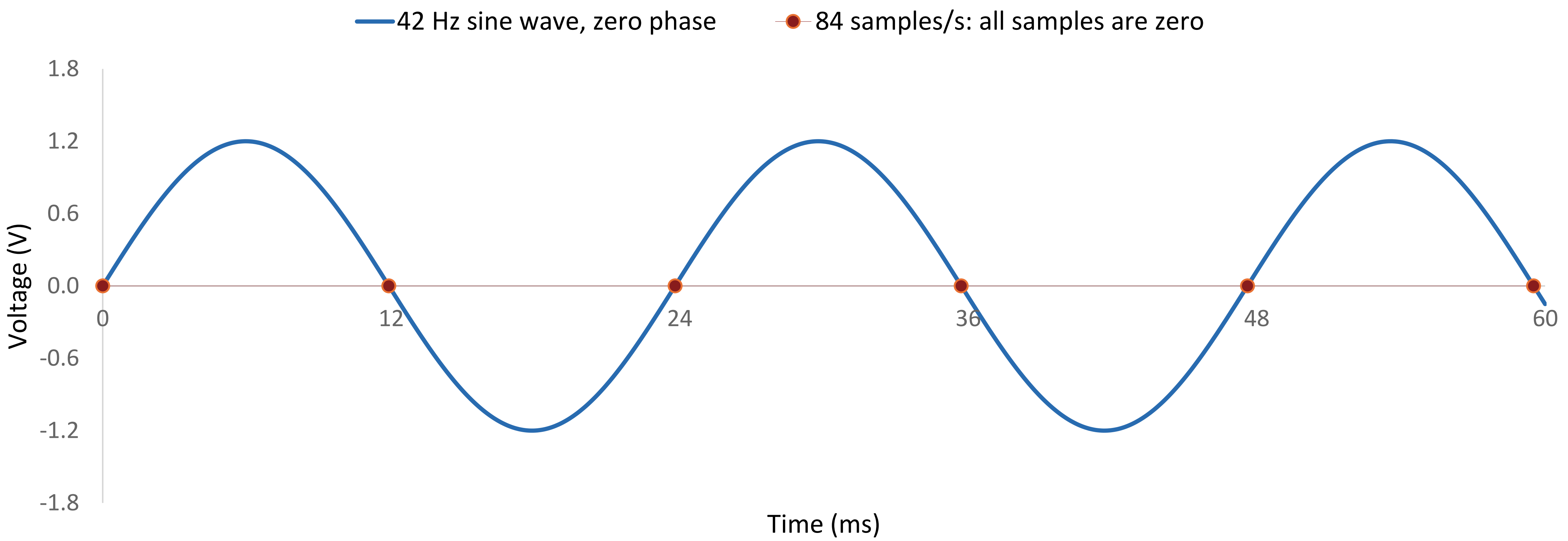
Nyquist frequency: $f_N = f_s / 2$.

At $f_s = 126$ samples/s: $f_N = 63$ Hz.

Both 18 Hz and 42 Hz are below 63 Hz.

The same condition is $f_{max} < f_N$.

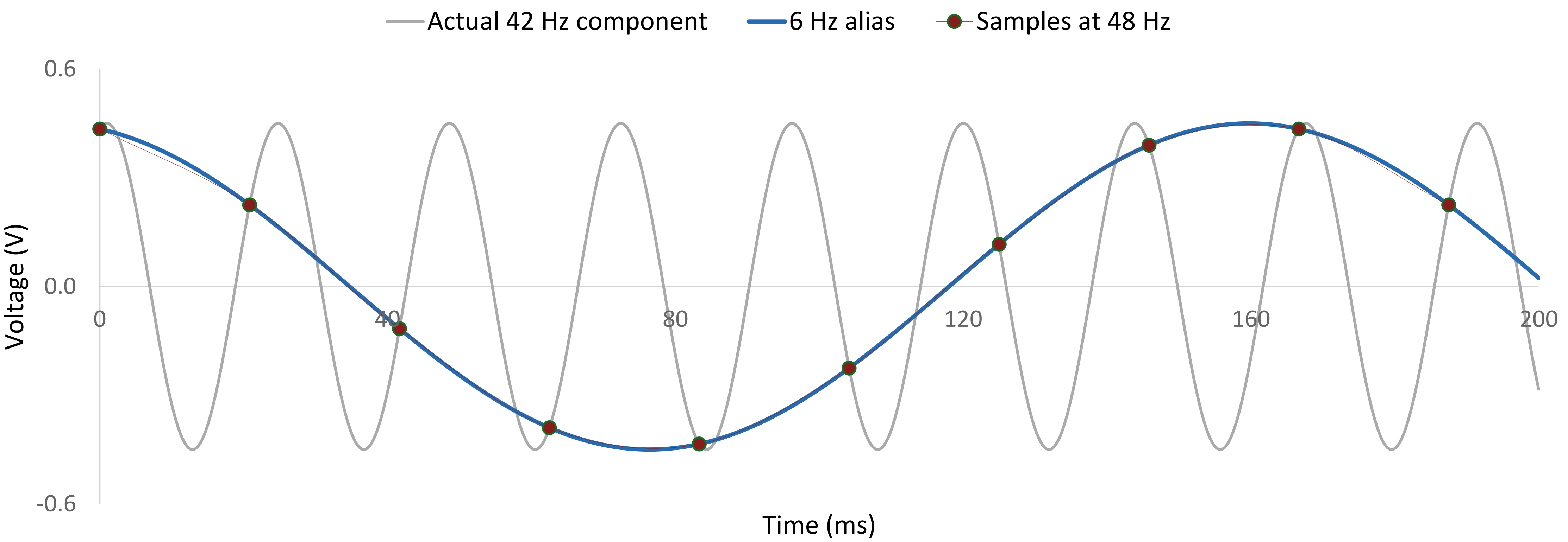
Exactly two samples per cycle can fail



At f_s = 2 f_signal, this starting phase makes every reading zero.

Use f_s > 2 f_max and leave margin for real measurements.

Aliasing: two waves fit the same samples



Both waves give the same sample values at fs = 48 Hz

42 Hz folds to |42 − 48| = 6 Hz. The signal itself has not changed.

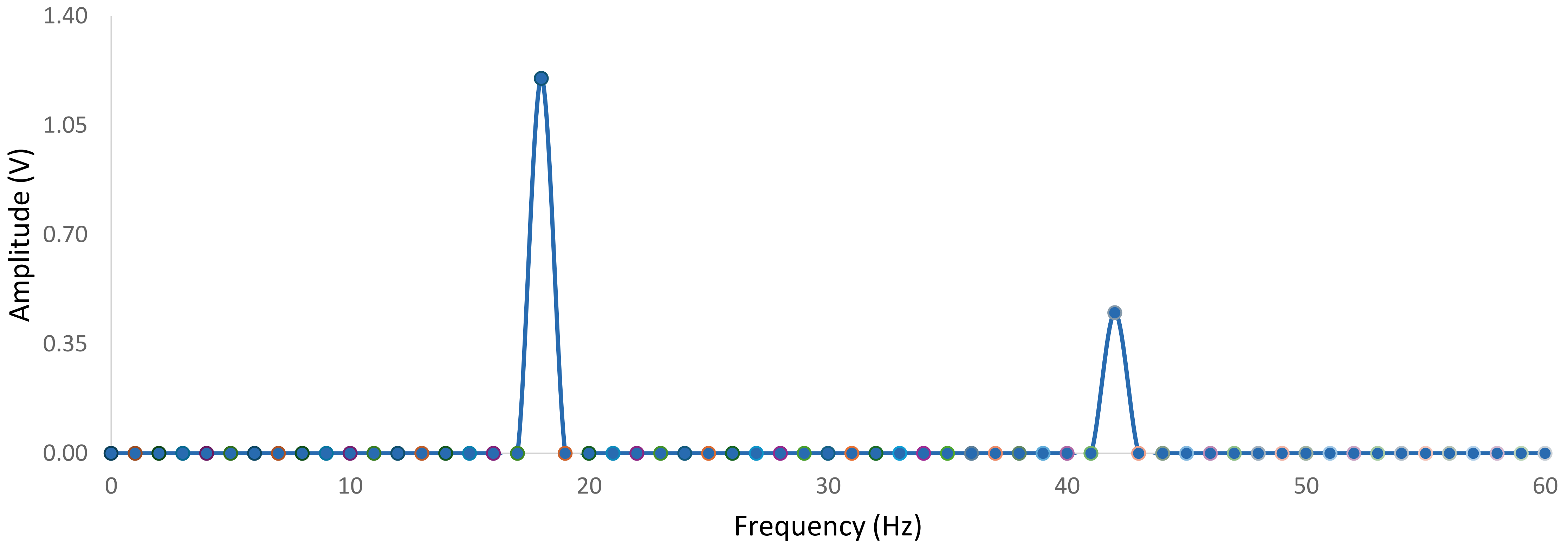
Predict an alias before measuring

- Choose an integer k so $f_{alias} = |f_{signal} - k f_s|$ lies in $[0, f_s/2]$.
- For a 42 Hz tone at 48 samples/s: $|42 - 48| = 6$ Hz.
- For a pure tone sampled at its own frequency, every reading has the same phase.
- A longer record or an FFT cannot recover information lost through aliasing.

Fourier series, transform, DFT, and FFT

- Fourier series: periodic waveform = sum of harmonic sinusoids.
- Fourier transform: frequency description of suitable continuous signals.
- DFT: frequency coefficients of N stored samples, X[k].
- FFT: an efficient algorithm for computing the DFT.

A waveform and a spectrum answer different questions



18 Hz peak: 1.20 V. 42 Hz peak: 0.45 V. Same samples, another view.

Calculated from 336 samples in 1 s. Both tones complete whole cycles; no taper in this example.

Reading an FFT peak

HORIZONTAL POSITION

The x-axis is frequency in Hz.

A peak at 18 Hz means 18 cycles/s.

This axis is not time.

It is not the recorder sampling rate.

VERTICAL HEIGHT

Here the y-axis is amplitude in volts.

The 18 Hz peak is about 1.20 V.

This is peak amplitude, not RMS.

Its single-tone RMS is $1.20/\sqrt{2}$ V.

Record duration and FFT frequency bins

THREE ACQUISITION QUANTITIES

N = number of stored samples.

f_s = samples per second.

$T_{record} = N / f_s$, in seconds.

T_{record} is not the period of one wave.

THE FFT FREQUENCY GRID

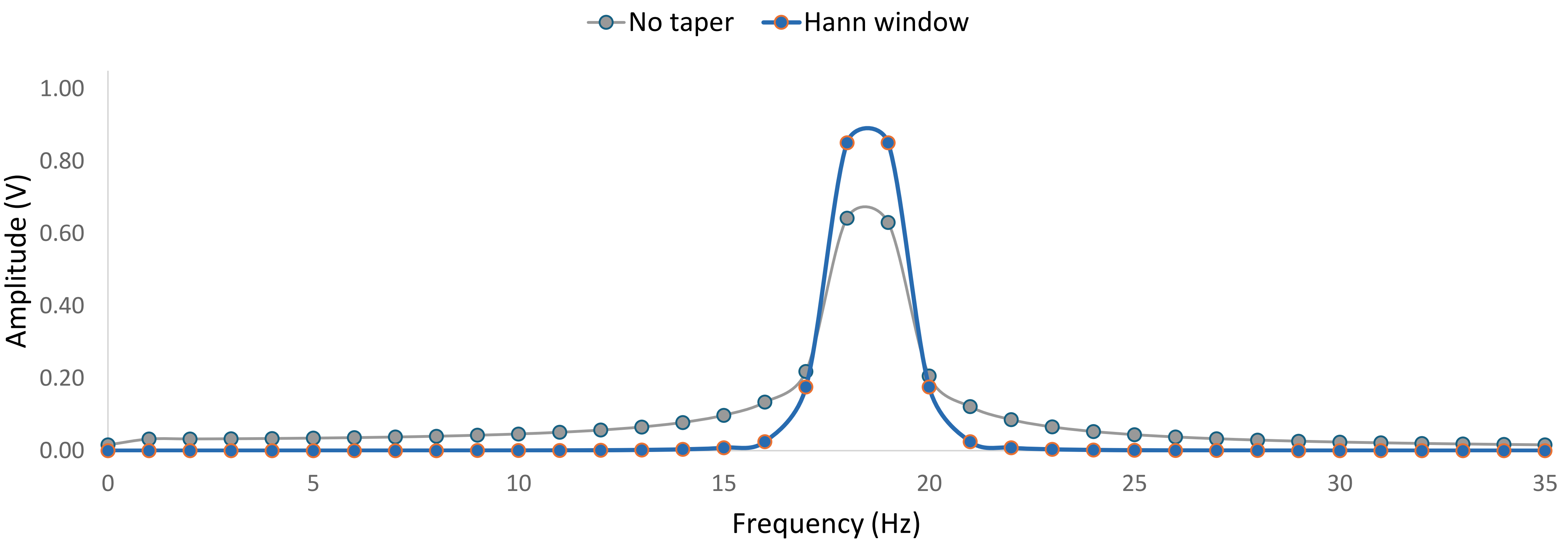
A bin is one frequency checked by the FFT.

Bin spacing: $\Delta f = f_s / N = 1/T_{record}$.

$T_{record} = 1$ s gives bins 1 Hz apart.

$T_{record} = 0.25$ s gives bins 4 Hz apart.

Why a peak can spread over several bins



One 18.5 Hz tone lies between 1 Hz bins. Neighboring bins are not extra tones.

Same 1 s record in both curves. The Hann window reduces distant leakage but widens the peak.

Windowing and filtering have different jobs

- Hann window: taper record ends; reduce distant leakage, but widen peaks.
- Amplitude scaling includes window gain; off-bin peaks still need care.
- Anti-alias filter: attenuate unwanted high frequencies before downsampling.
- Neither a window nor later smoothing can reverse an existing alias.

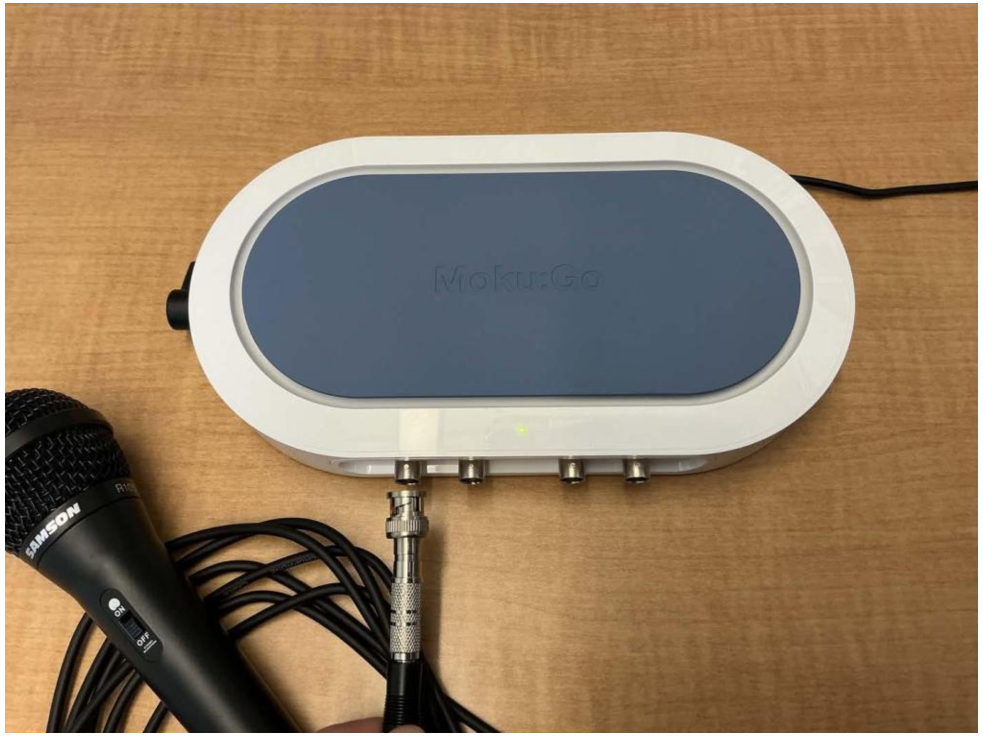
Lab 1: connect the actual equipment

Microphone → active Moku:Go INPUT → Data Logger → MAT file.

Use an input BNC connector; push and twist the connector to lock.

Select the Moku with the matching serial number in the desktop app.

Enable the connected input and disable the unused input.



Equipment photograph from the supplied Lab 1 slides

Prepare three different sound sources

Tuning fork: read its stamped frequency; excite it and place it near the microphone.

Whistle: hold a steady tone as consistently as possible.

Finger snap: capture the brief event and its decay inside the record.

Keep distance, input settings, and background conditions consistent.

Equipment photograph from the supplied Lab 1 slides

What does f mean in the laboratory task?

- For the tuning fork, f_ref is its labeled fundamental frequency.
- For a steady whistle, estimate f_ref from an adequately sampled pilot record.
- A snap has no single fundamental: identify and state a reference frequency.
- f_ref sets the comparison rates; f_max sets the anti-alias requirement.

The recording matrix: at least 15 records

- For EACH of at least three sounds, record at 100f_ref, 10f_ref, 3f_ref, 1.4f_ref, and 1f_ref.
- Every record must last at least 1 second. Save the actual rate and duration.
- For a 256 Hz fork: 25,600; 2,560; 768; 358.4; and 256 samples/s.
- The last two settings deliberately undersample the reference tone.

Set Data Logger before pressing Record

- Acquisition: enter Rate and record the selected mode, e.g. Precision.
- Logging: choose Immediate or Delayed and set Duration ≥ 1 s.
- Filename prefix: include source, reference frequency, rate, and trial.
- Start recording; inspect the waveform for clipping, silence, and missed events.

Export MAT files and verify one immediately

- Open File Manager; select recordings → Convert to MATLAB → Download.
- Keep the exported time column and voltage column together.
- Plot one file immediately and calculate the actual sample interval.
- Back up the data; do not leave with only a screenshot or an empty file.

Replay at the recorded rate

- Playback must use the original f_s, or an explicitly resampled audio signal.
- N is a sample count; it equals f_s numerically only for a 1 s record.
- Changing playback rate changes pitch and duration.
- Compare what you hear with the waveform and spectrum; note device limits.

Analyze: waveform, statistics, and spectrum

- Waveform: same time scale, labeled volts and seconds; show sample markers.
- Statistics: mean and RMS over a stated interval, with consistent gain.
- Spectrum: report f_s, N, window, FFT-bin spacing, and main peak locations.
- Compare each low-rate case with the high-rate reference and the predicted alias.

Before leaving: show evidence and save it

- Show the GTA one valid high-rate record and its spectrum.
- For the tuning fork, identify a clear peak near the labeled frequency.
- Confirm all sound/rate cases, metadata, MAT files, and backups are present.
- Report: method, results, comparison, discussion, conclusions, and references.

Pre-lab: rehearse the same measurement decisions

- A prepared 320 Hz microphone-voltage model replaces the physical source.
- Use the lab’s five rates: 100f, 10f, 3f, 1.4f, and 1f.
- Predict, run, compare waveforms and FFTs, then justify recording settings.
- All code is supplied; submit calculations, evidence, and explanations.

Final check: explain your measurement

- What is a physical signal frequency, and what is the recorder sample rate?
- Why can a low-rate waveform or FFT show a false frequency?
- What do you save before leaving the laboratory?
- Pre-labs normally appear Monday; submit before your own lab section begins.